Grade 11, Term 2, Lesson 1, Criterion C1

Parameters and sample statistics

C1. Distinguishes population parameters from sample statistics by interpreting and comparing the notation for population and sample means, standard deviations, and proportions in economic and financial contexts.

Highlight legend

Sample (green)	Population (red)

Highlight legend

Sample (green)	Population (red)
Sample (selected units)	Population (all units of interest)

Highlight legend

Sample (green)	Population (red)
Sample (selected units)	Population (all units of interest)
Sample mean ($\bar{x}$)	Population mean (μ)

Highlight legend

Sample (green)	Population (red)
Sample (selected units)	Population (all units of interest)
Sample mean ($\bar{x}$)	Population mean (μ)
Sample standard deviation (s)	Population standard deviation (σ)

Highlight legend

Sample (green)	Population (red)
Sample (selected units)	Population (all units of interest)
Sample mean ($\bar{x}$)	Population mean (μ)
Sample standard deviation (s)	Population standard deviation (σ)
Sample proportion ($\hat{p}$)	Population proportion (p)

Which electricity bills are we studying?

A utility studies the **amount charged per bill**.

Predict. Which bills might the utility want to describe? Which bills could it use to estimate the mean charge?

Which electricity bills are we studying?

A utility studies the **amount charged per bill**.

Predict. Which bills might the utility want to describe? Which bills could it use to estimate the mean charge?

Target population. All May electricity bills | for small shops | in one district.

Which electricity bills are we studying?

A utility studies the **amount charged per bill**.

Predict. Which bills might the utility want to describe? Which bills could it use to estimate the mean charge?

Target population. All May electricity bills for small shops in one district.

Sample. 40 randomly selected bills from that list.

From electricity bills to a mean

Predict. Which group does each mean describe?

Population parameter	Sample statistic

From electricity bills to a mean

Predict. Which group does each mean describe?

Population parameter	Sample statistic
μ is the unknown mean charge across all May electricity bills for small shops in the district.	

From electricity bills to a mean

Predict. Which group does each mean describe?

Population parameter	Sample statistic
μ is the unknown mean charge across all May electricity bills for small shops in the district.	$\bar{x} = 210$ thousand COP is the mean charge across the 40 selected bills.

From electricity bills to a mean

Predict. Which group does each mean describe?

Population parameter	Sample statistic
μ is the unknown mean charge across all May electricity bills for small shops in the district.	$\bar{x} = 210$ thousand COP is the mean charge across the 40 selected bills.

The population mean μ is fixed for this population. The sample mean $\bar{x}$ is calculated from the 40 bills and estimates that unknown mean.

Quick check. Redraw or redefine?

Predict. Which population mean does each draw estimate?

New draw
40 bills from the original May list
40 bills from the combined May list for small shops in two districts

Quick check. Redraw or redefine?

Predict. Which population mean does each draw estimate?

New draw	Suggested answer
40 bills from the original May list	A fresh draw can give a different $\bar{x}_{1\mathrm{D}}$ while estimating the same $\mu_{1\mathrm{D}}$, the mean charge for the original district.
40 bills from the combined May list for small shops in two districts	

Quick check. Redraw or redefine?

Predict. Which population mean does each draw estimate?

New draw	Suggested answer
40 bills from the original May list	A fresh draw can give a different $\bar{x}_{1D}$ while estimating the same μ_{1D}, the mean charge for the original district.
40 bills from the combined May list for small shops in two districts	This draw's mean $\bar{x}_{2D}$ estimates μ_{2D}, the mean charge across both districts.

Quick check. Redraw or redefine?

Predict. Which population mean does each draw estimate?

New draw	Suggested answer
40 bills from the original May list	A fresh draw can give a different $\bar{x}_{1D}$ while estimating the same μ_{1D}, the mean charge for the original district.
40 bills from the combined May list for small shops in two districts	This draw's mean $\bar{x}_{2D}$ estimates μ_{2D}, the mean charge across both districts.

The two population means could have the same numerical value.

A market association's sales survey (C1)

A market association wants to describe all 1,200 June-licensed stalls. It randomly selects 60 stalls.

In the survey, mean June sales revenue is 4.8 million COP and the standard deviation is 1.2 million COP. 18 sampled stalls accept digital payment.

Task 1. Which stalls make up the population? Which make up the sample?

A market association's sales survey (C1)

A market association wants to describe all 1,200 June-licensed stalls. It randomly selects 60 stalls.

In the survey, mean June sales revenue is 4.8 million COP and the standard deviation is 1.2 million COP. 18 sampled stalls accept digital payment.

Task 1. Which stalls make up the population? Which make up the sample?

Population. All 1,200 June-licensed stalls.

Sample. The 60 randomly selected stalls.

Match each quantity to its symbol

Task 1, continued. Which sample statistic estimates each population parameter? What does each symbol describe?

Population parameter	Sample statistic

Match each quantity to its symbol

Task 1, continued. Which sample statistic estimates each population parameter? What does each symbol describe?

Population parameter	Sample statistic
μ, mean June sales across all licensed stalls	$\bar{x}_1 = 4.8$ million COP, mean June sales across the 60

Match each quantity to its symbol

Task 1, continued. Which sample statistic estimates each population parameter? What does each symbol describe?

Population parameter	Sample statistic
μ, mean June sales across all licensed stalls	$\bar{x}_1 = 4.8$ million COP, mean June sales across the 60
σ, standard deviation of June sales across all licensed stalls	$s_1 = 1.2$ million COP, standard deviation of June sales across the 60

Match each quantity to its symbol

Task 1, continued. Which sample statistic estimates each population parameter? What does each symbol describe?

Population parameter	Sample statistic
μ, mean June sales across all licensed stalls	$\bar{x}_1 = 4.8$ million COP, mean June sales across the 60
σ, standard deviation of June sales across all licensed stalls	$s_1 = 1.2$ million COP, standard deviation of June sales across the 60
p, fraction of all June-licensed stalls accepting digital payment	$\hat{p}_1 = 0.30$, fraction accepting digital payment among the 60

The sample share accepting digital payment

Predict. What does each count in the fraction describe?

$$\hat{p}_1 = \frac{18}{60} = 0.30 = 30\%$$

The sample share accepting digital payment

Predict. What does each count in the fraction describe?

$$\hat{p}_1 = \frac{18}{60} = 0.30 = 30\%$$

Numerator. 18 sampled stalls accept digital payment.

The sample share accepting digital payment

Predict. What does each count in the fraction describe?

$$\hat{p}_1 = \frac{18}{60} = 0.30 = 30\%$$

Numerator. 18 sampled stalls accept digital payment.

Denominator. All 60 sampled stalls are counted.

The sample proportion is unitless.

A fresh draw from the June list

Population. All 1,200 June-licensed stalls. The association takes a fresh random sample from the same list to see how survey results vary across draws.

- Mean June sales revenue is 5.1 million COP.
- Standard deviation is 1.0 million COP.
- Digital-payment share is $0.25 = 25\%$.

Task 2. Revisit your answers to Task 1. What changes, what stays fixed, and which differences can you calculate?

Compare the two June draws

Population. All 1,200 June-licensed stalls.

Predict. Which sample statistics changed across the two draws?

Sample statistic	First draw	Second draw
June sales mean (million COP)		
June sales standard deviation (million COP)		
Digital-payment share (unitless)		

Compare the two June draws

Population. All 1,200 June-licensed stalls.

Predict. Which sample statistics changed across the two draws?

Sample statistic	First draw	Second draw
June sales mean (million COP)	$\bar{x}_1 = 4.8$	$\bar{x}_2 = 5.1$
June sales standard deviation (million COP)	$s_1 = 1.2$	$s_2 = 1.0$
Digital-payment share (unitless)	$\hat{p}_1 = 0.30$	$\hat{p}_2 = 0.25$

Compare the two June draws

Population. All 1,200 June-licensed stalls.

Predict. Which sample statistics changed across the two draws?

Sample statistic	First draw	Second draw
June sales mean (million COP)	$\bar{x}_1 = 4.8$	$\bar{x}_2 = 5.1$
June sales standard deviation (million COP)	$s_1 = 1.2$	$s_2 = 1.0$
Digital-payment share (unitless)	$\hat{p}_1 = 0.30$	$\hat{p}_2 = 0.25$

All three statistics differ in this fresh draw.

Second draw minus first draw

Predict. What are the three differences between sample statistics?

Sample statistic	Difference between draws
Mean	
Standard deviation	
Digital-payment share	

Predict. What stays fixed?

Second draw minus first draw

Predict. What are the three differences between sample statistics?

Sample statistic	Difference between draws
Mean	$\bar{x}_2 - \bar{x}_1 = 5.1 - 4.8 = +0.3$ million COP
Standard deviation	
Digital-payment share	

Predict. What stays fixed?

Second draw minus first draw

Predict. What are the three differences between sample statistics?

Sample statistic	Difference between draws
Mean	$\bar{x}_2 - \bar{x}_1 = 5.1 - 4.8 = +0.3$ million COP
Standard deviation	$s_2 - s_1 = 1.0 - 1.2 = -0.2$ million COP
Digital-payment share	

Predict. What stays fixed?

Second draw minus first draw

Predict. What are the three differences between sample statistics?

Sample statistic	Difference between draws
Mean	$\bar{x}_2 - \bar{x}_1 = 5.1 - 4.8 = +0.3$ million COP
Standard deviation	$s_2 - s_1 = 1.0 - 1.2 = -0.2$ million COP
Digital-payment share	$\hat{p}_2 - \hat{p}_1 = 25\% - 30\% = -5$ percentage points

Predict. What stays fixed?

Second draw minus first draw

Predict. What are the three differences between sample statistics?

Sample statistic	Difference between draws
Mean	$\bar{x}_2 - \bar{x}_1 = 5.1 - 4.8 = +0.3$ million COP
Standard deviation	$s_2 - s_1 = 1.0 - 1.2 = -0.2$ million COP
Digital-payment share	$\hat{p}_2 - \hat{p}_1 = 25\% - 30\% = -5$ percentage points

Predict. What stays fixed?

For the same June population, the June sales mean μ, June sales standard deviation σ, and digital-payment share p stay fixed. The statistics calculated from each sample estimate these parameters. Their exact population values remain unknown from the two draws.

Summary

Sample (green)	Population (red)
Sample (selected units)	Population (all units of interest)
Sample mean ($\bar{x}$)	Population mean (μ)
Sample standard deviation (s)	Population standard deviation (σ)
Sample proportion ($\hat{p}$)	Population proportion (p)